

The New York Times. In fact, in the languishing papers positing the birth of the universe, scientists were doubtful whether such radiation could ever be measured. As happens so often in discovery, those looking for evidence did not find it; those not looking for it found it and were hailed as discoverers.

We have a paradox. Not only have forecasters generally failed dismally to foresee the drastic changes brought about by unpredictable discoveries, but incremental change has turned out to be generally slower than forecasters expected. When a new technology emerges, we either grossly underestimate or severely overestimate its importance. Thomas Watson, the founder of IBM, once predicted that there would be no need for more than just a handful of computers.

That the reader of this book is probably reading these lines not on a screen but in the pages of that anachronistic device, the book, would seem quite an aberration to certain pundits of the “digital revolution.” That you are reading them in archaic, messy, and inconsistent English, French, or Swahili, instead of in Esperanto, defies the predictions of half a century ago that the world would soon be communicating in a logical, unambiguous, and Platonically designed lingua franca. Likewise, we are not spending long weekends in space stations as was universally predicted three decades ago. In an example of corporate arrogance, after the first moon landing the now-defunct airline Pan Am took advance bookings for round-trips between earth and the moon. Nice prediction, except that the company failed to foresee that it would be out of business not long after.

A Solution Waiting for a Problem

Engineers tend to develop tools for the pleasure of developing tools, not to induce nature to yield its secrets. It so happens that *some* of these tools bring us more knowledge; because of the silent evidence effect, we forget to consider tools that accomplished nothing but keeping engineers off the streets. Tools lead to unexpected discoveries, which themselves lead to other unexpected discoveries. But rarely do our tools seem to work as intended; it is only the engineer’s gusto and love for the building of toys and machines that contribute to the augmentation of our knowledge. Knowledge does not progress from tools designed to verify or help theories, but rather the opposite. The computer was not built to allow us to develop new, visual, geometric mathematics, but for some other purpose. It happened to allow us to discover mathematical objects that few cared to look for. Nor was the computer invented to let you chat with your friends in Siberia, but it has caused some long-distance relationships to bloom. As an essayist, I can attest that the Internet has helped me to spread my ideas by bypassing journalists.